(2) $B = 0$ and either A or $C = 0$, the general equation yields a parabola under study, at this level.

(3) $A \neq C$ and A and C are of the same sign the general equation yields an ellipse.

(4) $A \neq C$ and A and C are of opposite signs the general equation yields a hyperbola

(5) $A = C$ and $B = D = E = F = 0$, the general equation yields a point $x^2 + y^2 = 0$.

(6) $A = C = F$ and $B = D = E = 0$, the general equation yields an empty set $x^2 + y^2 + 1 = 0$, as there is no real solution.

(7) $A \neq 0$ or $C \neq 0$ and others are zeros, the general equation yield coordinate axes.

(8) $A = -C$ and rests are zero, the general equation yields a pair of lines $x^2 - y^2 = 0$.

ICT CORNER

https://ggbm.at/vchq92pg or Scan the QR Code

Open the Browser, type the URL Link given below (or) Scan the QR code. GeoGebra work book named "12th Standard Mathematics" will open. In the left side of the work book there are many chapters related to your text book. Click on the chapter named "Two Dimensional Analytical Geometry-II". You can see several work sheets related to the chapter. Select the work sheet "Conic Tracing"

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